

Note 13.1 Curves in Space and Their Tangents

Definitions

To describe a particle's coordinates as functions defined on I :

$$x = f(t), y = g(t), z = h(t), t \in I$$

And the points $(x, y, z) = (f(t), g(t), h(t)), t \in I$ make up the **curve** in space and we call it the **path** of the particle. And the upper equations and interval **parametrize** the curve.

A curve in space can be also represented in vector form,

$$\mathbf{r}(t) = \overrightarrow{OP} = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

The vector is from the origin to the particle's **position** $P(f(t), g(t), h(t))$ at time t . And it is the particle's **position vector**. The functions f, g, h are the **component functions (components)** of the position vector. The particle's path as the **curve traced by $\mathbf{r}$** during the time interval I . And the function is called **vector-valued function** or **vector function**.

Real-valued functions are called **scalar functions**.

Limits and Continuity

Definition of Limit

Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function with domain D and $\mathbf{L}$ a vector. Then we say that $\mathbf{r}$ has **limit $\mathbf{L}$** as $t \rightarrow t_0$ and write

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$$

if for every $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all $t \in D$,

$$0 < |t - t_0| < \delta \implies |\mathbf{r}(t) - \mathbf{L}| < \epsilon$$

And if $\mathbf{L} = L_1\mathbf{i} + L_2\mathbf{j} + L_3\mathbf{k}$, then it can be shown that $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$ when

$$\lim_{t \rightarrow t_0} f(t) = L_1, \lim_{t \rightarrow t_0} g(t) = L_2, \lim_{t \rightarrow t_0} h(t) = L_3$$

So the equation

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \left(\lim_{t \rightarrow t_0} f(t) \right) \mathbf{i} + \left(\lim_{t \rightarrow t_0} g(t) \right) \mathbf{j} + \left(\lim_{t \rightarrow t_0} h(t) \right) \mathbf{k}$$

provides a practical way to calculate limits.

Definition of Continuity

A vector function $\mathbf{r}(t)$ is **continuous** at $t = t_0$ in its domain if $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$ and the function is **continuous** if it is continuous at every point in its domain.

Derivatives and Motion

Definition of Derivative

The vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ has a **derivative(is differentiable)** at t if f, g, h have derivatives at t . And the derivative is

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{df}{dt}\mathbf{i} + \frac{dg}{dt}\mathbf{j} + \frac{dh}{dt}\mathbf{k}$$

And the vector function is **differentiable** if it is differentiable at every point of its domain. And the curve is **smooth** if $d\mathbf{r}/dt$ is continuous and never $\mathbf{0}$. The vector $\mathbf{v}'(t)$ is the **tangent** of the curve at t and the tangent line to the curve is the line through the point $(f(t_0), g(t_0), h(t_0))$ with the tangent $\mathbf{r}'(t_0)$.

Definition of Motion

If $\mathbf{r}$ is the position vector of a particle moving along a smooth curve in space, then

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

is the particle's **velocity vector**, tangent to the curve. At any time t , the direction of $\mathbf{v}$ is the **direction of motion**. The magnitude of $\mathbf{v}$ is **speed**. And the derivative $\mathbf{a} = d\mathbf{v}/dt$ is the particle's **acceleration vector**.

Differentiable Rules

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions at t , $\mathbf{C}$ a constant vector and c any scalar, f any differentiable scalar function.

Rule	Formula
Constant Function Rule	$\frac{d}{dt}\mathbf{C} = \mathbf{0}$
Scalar Multiple Rules	$\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$

Rule	Formula
	$\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
Sum Rule	$\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
Dot Product Rule	$\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
Cross Product Rule	$\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
Chain Rule	$\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

The proof can be done by the definitions.

Vector Functions of Constant Length

If $\mathbf{r}$ is a differentiable vector function of t of constant length, then

$$\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$$

That is, the position vector $\mathbf{r}$ is always perpendicular to its velocity vector.

For example, the particle moving on a sphere centered at the origin with radius c have

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = c^2$$

take derivatives of both sides

$$2\mathbf{r}'(t) \cdot \mathbf{r}(t) = 0.$$

That is

$$\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$$

Exercises